

Testing and Evaluation of Agentic Systems

Agenda

- Human-in-the-Loop
- Evaluating Multi-Agent Systems

Let's begin the discussion by answering a few questions.

Testing and Evaluation of Agentic Systems Quiz

What is the primary difference between evaluating a single LLM and evaluating a multi-agent system?

A

Single LLMs are deterministic; agents are not

B

Multi-agent systems require evaluation of workflow, coordination, and tool use

C

Single LLMs do not produce outputs

D

Agents do not use prompts

Testing and Evaluation of Agentic Systems Quiz

What is the primary difference between evaluating a single LLM and evaluating a multi-agent system?

A

Single LLMs are deterministic; agents are not

B

Multi-agent systems require evaluation of workflow, coordination, and tool use

C

Single LLMs do not produce outputs

D

Agents do not use prompts

Evaluating Multi-Agent Systems

Evaluation Dimension	What It Measures	Why It Matters
Reasoning Coherence	Logical consistency across multi-step reasoning	Prevents contradictory or drifting arguments
Tool Call Accuracy	Correct tool selection, correct parameters, successful execution	Ensures external functions are used properly
Workflow Coordination	Proper handoff between agents, correct sequencing	Prevents cascading failures and miscommunication
State & Memory Consistency	Stability of context, memory updates, and goal alignment over time	Prevents drift, infinite loops, and overwritten goals

Testing and Evaluation of Agentic Systems Quiz

Why is human-in-the-loop recommended for high-risk AI systems?

A

To make the system faster

B

To ensure a qualified human reviews and validates decisions before they affect critical real-world outcomes

C

Because LLMs cannot generate text properly

D

To reduce token usage

Testing and Evaluation of Agentic Systems Quiz

Why is human-in-the-loop recommended for high-risk AI systems?

A

To make the system faster

B

To ensure a qualified human reviews and validates decisions before they affect critical real-world outcomes

C

Because LLMs cannot generate text properly

D

To reduce token usage

Human-in-the-Loop

Human Oversight Type	Where It Appears in Workflow	Level of Automation	Purpose
Human-in-the-Loop	During execution (before final output is acted upon)	Low automation	Human reviews, approves, or modifies agent decisions before real-world impact
Human-on-the-Loop	After execution but during monitoring	Medium automation	System runs autonomously, but humans supervise and intervene if anomalies occur
Human-over-the-Loop	Periodic evaluation after deployment	High automation	Humans audit logs, performance metrics, and decision trails
Human-out-of-the-Loop	No routine human intervention	Very high automation	System operates independently with pre-defined safeguards



Power Ahead!

